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Duration	9 mins 11 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Bubble Sort
- b. Merge Sort
- c. Quick Sort
- d. Insertion Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort

b. Selection sort

c. Merge sort

d. Quick sort

The correct answer is:
Merge sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass

b. It is less efficient than bubble sort

c. It is based on the divide-and-conquer approach

d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 40,80,90,150,200,300,400,470,70,120

c. 80,150,200,470,40,90,300,400,70,120

d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sort()
b. arrange()
c. sorted()
d. order()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Decreases the efficiency of algorithms
c. Provides a basis for other algorithms to work efficiently
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Merge Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It allows for faster searching
c. It has no effect on the search operation
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data in a single pass
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Quick Sort
c. Heap Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It only works with integer arrays
c. It requires external libraries
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It only works on small datasets
c. It avoids the need for recursion
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data